



اَوْنُوْ سِيْطِيْ تِيْكَوْ لُوْ كِيْ مَارَا
UNIVERSITI
TEKNOLOGI
MARA

**OBJECT ORIENTED PROGRAMMING
(CSC186)**

GROUP ASSIGNMENT

Prepared By: **ASMA' SALSABILA BINTI MAUSENG SAMA-
AE**

2019284386

MICHELLE LIM AK LAUNCESTON DAVE

2019206024

NUR SYAQEERA BINTI SHAHARIZAN

2019433956

ALYA NAZIRAH BINTI MOHD SUFIAN

2019420962

NUR IRDINA IZZATI BINTI ZULKIFLY

2019242886

KCS1102G

Prepared For: **SIR MUHAMMAD KHAIRUL ZHARIF BIN NOR
A'ZAM**

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ONLINE BANKING SYSTEM: FUND TRANSFER

1. INTRODUCTION

The “Online Banking System” characterized as the integrated distribution of modern and conventional banking products and services directly to customers through online means, it involves programs that allow customers of financial institutions, companies to access accounts, to transact companies or to obtain details on financial services via a public or private network like the internet.

This group consists of five members, Salsa, Michelle, Syaqeera, Alya and Irdina. Our project was done by dividing the tasks to each one of us. The UML diagram was completed by Alya and Dina. The pseudocode documentation was done by Salsa and Syaqeera. The coding documentation was created by Michelle. The report writing task was done by all of the team members. After completing our given tasks, we have discussed our works altogether for more better ideas and to detect any blemishes in order to produce a perfect finishing.

We have faced some difficulties in coding the codes but with some opinions and knowledges from team members, the last product of our coding seemed better and arranged. For the accuracy, we are not entirely sure of it but we have exerted the project the best as we could. In conclusion, to succeed this project, all of the team members have shown such a great enthusiasm and responsibility. This project could not have been done without this kind of teamwork and encouragement.

2. OBJECTIVE

The key aim of the “Online Banking System” project is to handle account data and transaction. It manages all the deposits, balance and online banking records. The project is solely designed for administrative purposes and further access is given only to the administrator. The initiative aims at developing a system to of the administrative work for the administration of deposits, bank, balance sheets, customers. This maintains track of all client data, transactions and online banking. It also provides the searching facilities based on various factors. Such as accounts, transaction and balance. It tracks all the information of customer, balance and transaction. Shows the information and description of the accounts also one of the objectives of this project.

3. USE CASE ANALYSIS

3.1 Identified use cases

1. Login

An action performed by the customer when customer want to get into their bank account.

2. Verify Password

An action performed by the system to make sure that the customer is the owner of the bank account.

3. Display Login Error

An action performed by the system if the customer filled the login with wrong details.

4. Check Balance

An action performed by the customer to check their account balance.

5. Transfer Funds

An action performed by the customer if they want to transfer funds to other bank accounts.

6. Verify Sufficient Funds

An action performed by the system to make sure customer have enough funds before transferring funds or make payment.

7. Fill details

An action performed by the customer if they want to transfer funds to other bank accounts.

8. Verify with One-Time Password

An action performed by the customer to fill in one-time password for a successful transaction.

3.2 Identified Actors

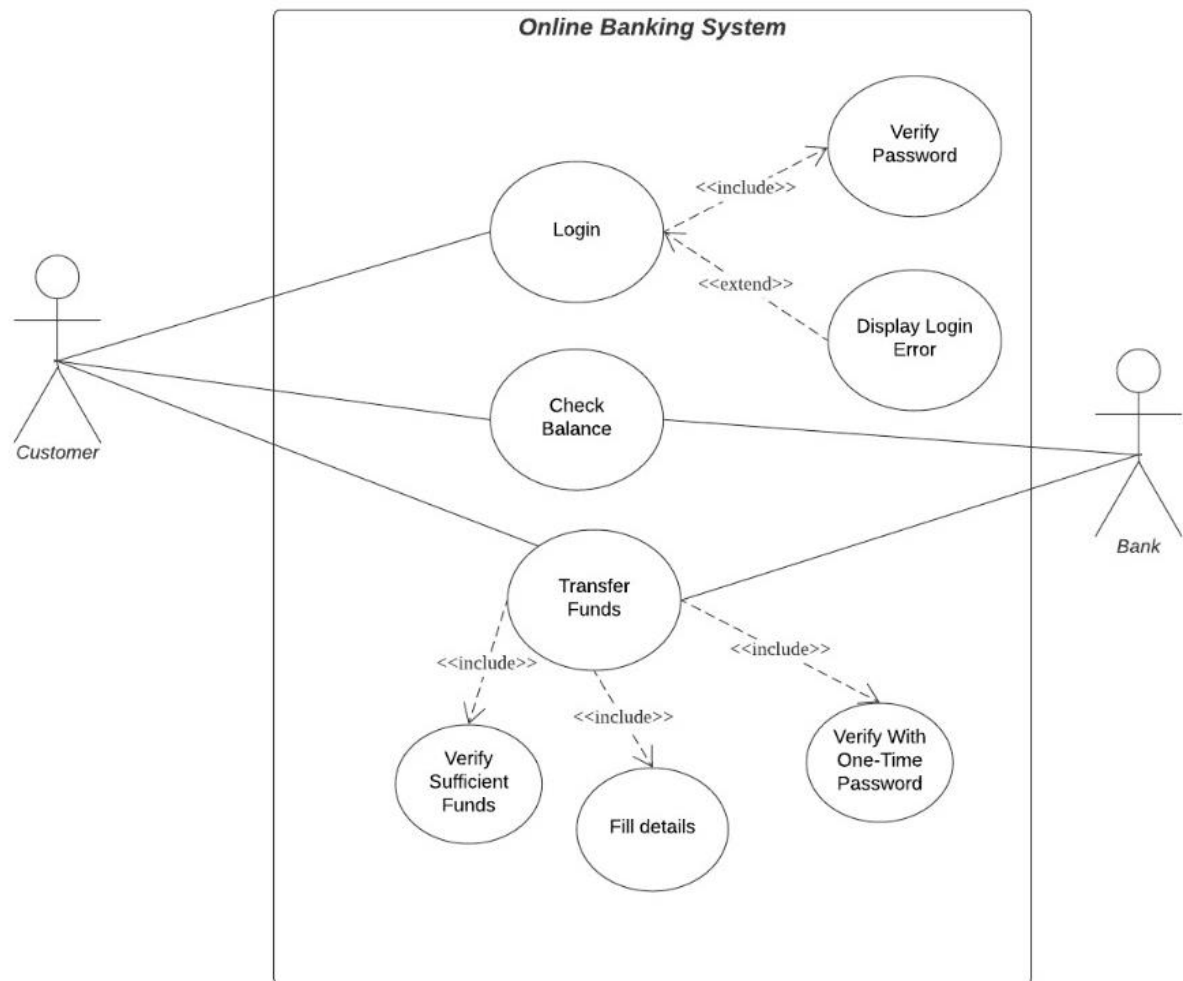
1. Customer

The customer can login, check their balance account and transfer funds by using the online banking system. Customer must first verify their password before they get to login into their bank accounts. They also need to fill the details of other bank accounts and one-time password given to their phone numbers to complete a transaction.

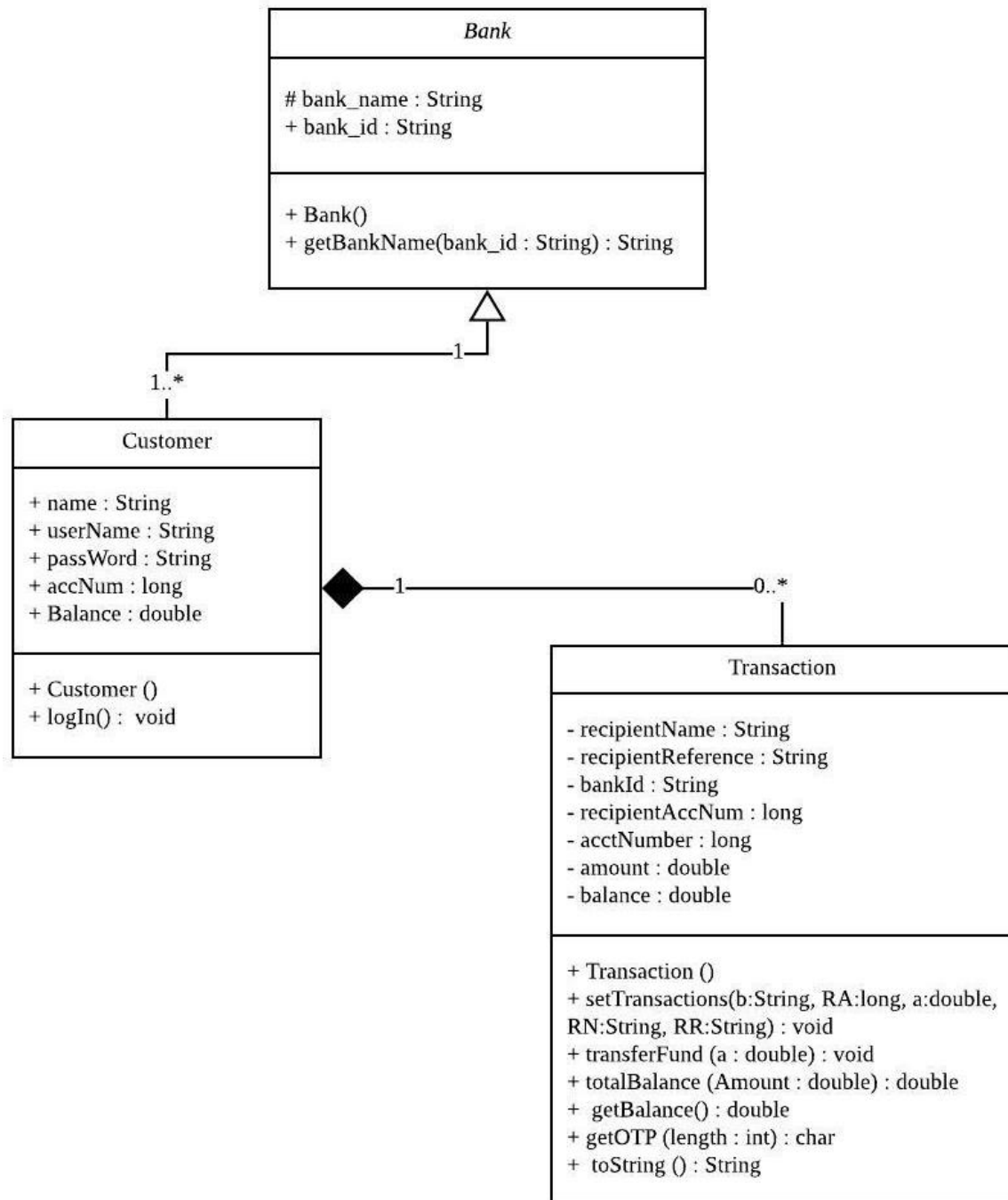
2. Bank

The bank should provide the correct amount of the customer's balance in their bank account every time they want to check their balance. Similarly, when a customer wants to transfer funds, the bank is going to follow through with those transaction.

4. USE CASE DIAGRAM



5. CLASS DIAGRAM



6. CODING DOCUMENTATION

6.1 Main class

```
package com.onlineBanking;
import java.util.Scanner;

public class onlineBankingSystem{

    public static void main(String[] args) {
        String input02 = "null", bankID, recipient_Name, recipient_Reference;
        int input01;
        double balance, Amount;
        long recipient_AccNum;

        Customer a = new Customer();
        Transaction b = new Transaction ();

        Scanner input = new Scanner (System.in);
        Scanner input2 = new Scanner (System.in);
        Scanner input3 = new Scanner (System.in);

        a.logIn();

        if (a.userName.equals("user123") && a.passWord.equals("abc123")) {

            System.out.println("\n\nYou've successfully logged in to this online banking system");

            while (!input02.equals("N") && !input02.equals("n")) {

                // main menu
                System.out.println("\nWelcome!");
                System.out.println("\n\t1 - Check Balance \n\t2 - Transfer Fund");

                System.out.print("\nPlease enter your option: ");
                input01 = input.nextInt();

                //option 1
                if(input01 == 1) {

                    balance = b.getBalance();
                    System.out.println("\nYour account balance: RM " + balance);

                }

                // option 2
                else if(input01 == 2) {
                    System.out.println("\n\tTRANSACTION DETAILS - Please fill up everything before making an online transfer");
                    System.out.println("\nBank ID available => \n\tBSN - BANK SIMPANAN NASIONAL\n\tMBB - MAYBANK\n\tTALM - ALLIANCE BANK");
                    System.out.println("\tBIB - BANK ISLAM\n\tAMB - AmBANK\n\tCIMB - CIMB BANK");

                    System.out.print("\nPlease enter Bank ID : ");
                    bankID = input2.nextLine().trim().toUpperCase();
                    System.out.print("Please enter recipient account number: ");
                    recipient_AccNum = input2.nextLong();
                    System.out.print("Please enter amount : RM ");
                    Amount = input2.nextDouble();
                }
            }
        }
    }
}
```

```

        System.out.print("Please enter recipient name: ");
        recipient_Name = input3.nextLine().toUpperCase();
        System.out.print("Please enter recipient reference: ");
        recipient_Reference = input3.nextLine();

        b.setTransactions(bankID,recipient_AccNum,Amount,recipient_Name,recipient_Reference);
    }

    System.out.print("\nDo you wish to continue? (Y/N): ");
    input02 = input.next();
    System.out.println("-----");
}

else {
    System.out.println("\n\nOh no! incorrect username or password. Please try again later");
}

}

}

```


6.2 Bank class

```
package com.onlineBanking;

public class Bank {

    protected static String bank_name;
    String bank_id;

    //default constructor
    public Bank () {

        bank_name = "Null";
        bank_id = "Null";

    }

    // getter method
    public String getBankName (String bank_id) {

        switch (bank_id.toUpperCase()) {

            case "BSN":
                bank_name = "BANK SIMPANAN NASIONAL";
                break;
            case "HBB":
                bank_name = "MAYBANK";
                break;
            case "ALM":
                bank_name = "ALLIANCE BANK";
                break;
            case "BIB":
                bank_name= "BANK ISLAM";
                break;
            case "AMB":
                bank_name= "AmBANK";
                break;
            case "CIMB":
                bank_name= "CIMB BANK";
                break;

        }

        return bank_name;

    }

}
```

6.3 Customer class

```
package com.onlineBanking;
import java.util.Scanner;

public class Customer extends Bank {
|
|
    public String name, userName, passWord;
    public long accNum;
    public double Balance;

    // default constructor for customer a
    public Customer () {

        bank_name = "BANK SIMPANAN NASIONAL";
        bank_id = "BSN";
        name="KIAH BINTI JALIL";
        Balance = 120_000.00;
        accNum = 1234567821;
        userName="user123";
        passWord="abc123";

    }

    public void logIn () {
        Scanner scanner = new Scanner (System.in);
        System.out.println("\nHello please enter username and password");
        System.out.print("\nUsername: "); // use System.out.print if u want the user input beside the question
        userName = scanner.next();
        System.out.print("Password: ");
        passWord = scanner.next() ;

    }
}
```

6.4 Transaction class

```
package com.onlineBanking;

import java.util.Random;
import java.util.Scanner;

public class Transaction {

    private String recipientName, bankId, recipientReference;
    private long recipientAccNum, acctNumber;
    private double amount, balance;

    Scanner Input = new Scanner (System.in);
    Scanner Input2 = new Scanner(System.in);
    Bank n = new Bank ();
    Customer a = new Customer ();

    // default constructor
    public Transaction () {

        balance = a.Balance;
        acctNumber=a.acctNum;
        recipientName ="null";
        recipientAccNum = 0;

    }

    // setter method
    public void setTransactions(String b, long RA, double a, String RN, String RR) {

        bankId = b;
        recipientAccNum = RA;
        amount = a;
        recipientName = RN;
        recipientReference = RR;

        transferFund(amount);

    }

    // this method is to ensure the customer is able to do online transfer
    public void transferFund (double a){

        if (a < 0) { // if transfer amount is negative

            System.out.println ();
            System.out.println ("Error! The amount that u tried transferring is invalid.");
            System.out.println ("Owner's account number: " + acctNumber);
            System.out.println ("Requested amount: RM " + a);

        }

        else
            if (a > balance){ // transfer amount exceeds balance

                System.out.println ();
                System.out.println ("Error! Insufficient funds available.");
                System.out.println ("Owner's account number: " + acctNumber);
```

```

        System.out.println ("Requested amount: RM " + a);
        System.out.println ("Available balance: RM " + balance);

    }

    else {

        System.out.println();
        System.out.println(getOTP( length: 4));
        totalBalance(a);
        System.out.println("You have successfully transferred the fund to " + recipientName);
        System.out.println(toString());
    }

}

// processor method - this method will calculate the total balance after customer transferred the funds
public double totalBalance (double Amount) { return balance = balance - Amount; }

// getter method
public double getBalance() { return balance ;}

// method for One Time Password - OTP / getter method
private static char [] getOTP(int length)throws IllegalArgumentException{

    System.out.print("One Time Password (OTP): ");
    String Number="012345678"; // the OTP numbers will be selected within these range of number
    char Otp[]=new char[length];
    Random getOtpNumber=new Random();
    for (int i =0;i < length; i++){
        Otp[i]=Number.charAt(getOtpNumber.nextInt(Number.length()));
    }
    return Otp;

}

//printer method
public String toString () {

    return "\n\tTRANSACTION DETAILS\n\n\tRecipient name: " +recipientName+
        "\n\tBank name: " + n.getBankName(bankId)+
        "\n\tAccount number: " + recipientAccNum+
        "\n\tAmount: RM "+amount+
        "\n\tRecipient reference: " +recipientReference+
        "\n\n\t\tTHANK YOU";

}

}

```

7. Sample input and output

7.1 Input for login

```
Hello please enter username and password

Username: user123
Password: abc123
```

7.2 Input and output for option 1

```
Welcome!

1 - Check Balance
2 - Transfer Fund

Please enter your option: 1

Your account balance: RM 119900.0

Do you wish to continue? (Y/N): n
-----

Process finished with exit code 0
```

After you done one transaction, you can choose either you want to continue with another transaction or not. If yes you need to click 'Y' or 'y' and you will continue another transaction if no click 'N' or 'n' and it will end.

```
You've successfully logged in to this online banking system

Welcome!

1 - Check Balance
2 - Transfer Fund

Please enter your option: 1

Your account balance: RM 120000.0

Do you wish to continue? (Y/N): y
-----
```

7.3 Input and output for option 2

```
Welcome!

1 - Check Balance
2 - Transfer Fund

Please enter your option: 2

TRANSACTION DETAILS - Please fill up everything before making an online transfer

Bank ID available =>
BSN - BANK SIMPANAN NASIONAL
MBB - MAYBANK
ALM - ALLIANCE BANK
BIB - BANK ISLAM
AMB - AmBANK
CIMB - CIMB BANK

Please enter Bank ID : bsn
Please enter recipient account number: 263736455364563
Please enter amount : RM 100.00
Please enter recipient name: Nur evita margie
Please enter recipient reference: study fees
```

7.4 Output transaction details after customer has successfully made their fund transfer

```
One Time Password (OTP): 6117
You have successfully transferred the fund to NUR EVITA MARGIE
```

TRANSACTION DETAILS

```
Recipient name: NUR EVITA MARGIE
Bank name: BANK SIMPANAN NASIONAL
Account number: 263736455364563
Amount: RM 100.0
Recipient reference: study fees
```

THANK YOU

7.5 Sample error message

Please enter your option: *2*

TRANSACTION DETAILS - Please fill up everything before making an online transfer

Bank ID available =>

BSN - BANK SIMPANAN NASIONAL

MBB - MAYBANK

ALM - ALLIANCE BANK

BIB - BANK ISLAM

AMB - AmBANK

CIMB - CIMB BANK

Please enter Bank ID : *cimb*

Please enter recipient account number: *55628354647782*

Please enter amount : RM *-8765.00*

Please enter recipient name: *oniel wagner*

Please enter recipient reference: *carousell*

Error! The amount that u tried transferring is invalid.

Owner's account number: 1234567821

Requested amount: RM -8765.0

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